

Math 330 Abstract Algebra Notes

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1 Introduction to Group Theory

1.1 Modular Arithmetic

Definition 1.1. Take $n > 0$ and $n \in \mathbb{Z}$. We say that $a, b \in \mathbb{Z}$ are *congruent modulo n* ($a \equiv b \pmod{n}$) if n divides $a - b$ ($n|a - b$) or if the remainder of n/a equals the remainder of n/b .

Here are some properties:

1. if $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then $a + c \equiv b + d \pmod{n}$.

Proof. $a \equiv b \pmod{n}$ implies that $n|a - b$ which implies that $\exists q \in \mathbb{Z}$ such that $a - b = nq$. The same can be said for $a \equiv b \pmod{n}$ for some $q' \in \mathbb{Z}$ such that $c - d = nq'$. Adding these equations together we have that $a - b + c - d = (a + c) - (b + d) = n(q + q')$. Since the integers are closed under addition, $q + q'$ is an integer and therefore $n|(a + c) - (b + d)$ and as a result, $a + c \equiv b + d \pmod{n}$. $\square$

2. if $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then $ac \equiv bd \pmod{n}$

Proof. $a \equiv b \pmod{n}$ implies that $n|a - b$ which implies that $\exists q \in \mathbb{Z}$ such that $a - b = nq$. The same can be said for $a \equiv b \pmod{n}$ for some $q' \in \mathbb{Z}$ such that $c - d = nq'$. We can write ac as

$$\begin{aligned} ac &= (b + nq)(d + nq') \\ &= bd + bnq' + dnq + n^2qq' \\ \Rightarrow ac - bd &= n(bq' + dq + nqq') \end{aligned}$$

and since everything in the parentheses are integers and the integers are closed under multiplication and addition, their result is an integer and therefore $n|ac - bd$ and $ac \equiv bd \pmod{n}$. $\square$

3. Congruence mod n is an equivalence relation

Proof. An equivalence relation is reflexive, symmetric, and transitive. To prove reflexivity, we show $a \equiv a \pmod{n} \rightarrow n|a - a \rightarrow n|0$ which is always true. To prove symmetry, we have that $a \equiv b \pmod{n}$ implies $(a - b) = nq$ which we can extract a negative sign such that $(b - a) = n(-q)$ and since $-q$ is also an integer, $n|b - a$ and therefore $b \equiv a \pmod{n}$. To prove transitivity, we say that $a \equiv b \pmod{n}$ implies $a - b = nq$ and $b \equiv c \pmod{n}$ implies $b - c = nr$. We now find that $a - c = b + nq + nr - b = n(q + r)$. Since the integers are closed under addition, $q + r \in \mathbb{Z}$ and therefore $n|a - c$ and $a \equiv c \pmod{n}$. $\square$

1.2 Introduction to Groups

Definition 1.2 (Group). A group is a set of elements G with a binary operation $*$ that assigns to every pair (g, h) of elements of G another element in G denoted $g * h$. Groups follow the group axioms:

1. Associativity

$$(g_1 * g_2) * g_3 = g_1 * (g_2 * g_3)$$

2. There exists an element $e \in G$ that is the identity such that

$$e * g_1 = g_1 * e = g_1$$

3. Every element has an inverse element denoted g^{-1} such that

$$g * g^{-1} = g^{-1} * g = e$$

4. The set of elements is closed under the group operation (although this is implied in the definition).

Example 1.3. Let us look at some examples

1. The set $(\mathbb{Z}, +)$. Addition is associative, the identity is 0, the inverse of each element g is $-g$.
2. The set $(\mathbb{R}_{>0}, \cdot)$ where $\cdot$ denotes regular multiplication. Regular multiplication is associative, the identity is 1, the inverse element is $\frac{1}{g}$.
3. The set $(\{-2, -1, 0, 1, 2\}, +)$. This fails due since it is not closed (take $1 + 2 = 3 \notin$ the set).
4. The set $(\{\frac{1}{2^n}\}, \cdot)$. This fails since there is no inverse.
5. Fix a positive integer n . Take the set $(\{0, 1, 2, \dots, n-1\}, + \text{ mod } n)$. The identity is 0, addition is associative. The inverse of an element $0 \leq a \leq n-1$ is $n-a$ since $(n-a) + a \text{ mod } n = n \text{ mod } n = 0$. This group is denoted by Fraleigh as $\mathbb{Z}_n$ but more generally denoted as $\mathbb{Z}/n\mathbb{Z}$.
6. Fix an $n \in \mathbb{N}$. We denote the group $\text{GL}_n(\mathbb{R})$ as the set of invertible $n \times n$ matrices with real entries with the operation of matrix multiplication. Matrix multiplication is associative, the identity is the $n \times n$ identity matrix I_n . The inverse of every element A is the inverse matrix of A . This group is also known as the *General Linear Group*.

1.3 Properties of Groups

Definition 1.4. The *order* of a group G , denoted $|G|$, is the cardinality of the group/set. If G is finite, then $|G| < \infty$.

Definition 1.5. If the group operation $*$ is commutative, then the group is *Abelian*.

Here are some properties:

1. The identity is unique

Proof. Suppose we have some $f \in G$ such that $f * g = g \ \forall g \in G$. Then if we compose f with the identity, we should get $f * e = e$. By axiom 2 of the group axioms, we must also have that $f * e = f$ and therefore we have that $f * e = e = f$. $\square$

2. Each inverse element is unique.

Proof. $(a')'a' = e \rightarrow (a')'a'a = ea \rightarrow (a')'e = a \rightarrow (a')' = a$ $\square$

3. Cancellation, that is if $a * b = c$ for some $a, b, c \in G$, then $b = a^{-1} * c$.

Proof. By group axiom 3, there exists an inverse a^{-1} . We can then left multiply by a^{-1} such that $a^{-1} * a * b = a^{-1} * c$ and by associativity, we then have $b = a^{-1} * c$. $\square$

The consequences of this is that we can make the following observations: $a * b = a * c$ implies that $b = c$ and $b * a = c * a$ implies $b = c$.

For finite groups, we can write a group table where we have the left argument along the rows and the right argument of the operation along the columns. Since Abelian groups are commutative, they have a symmetry across the diagonal. An interesting thing to note with these group tables is that every entry appears only once in each row and column.

Proof. If we have a repeated entry in a row, that implies $a * b = a * c$ but by cancellation, we have that $b = c$. For each column, this implies that $b * a = c * a$ which also implies $b = c$. $\square$

Example 1.6. Let us take a group of cardinality 2. We require that there exists an identity element so the set contains $\{e, a\}$ where a is some arbitrary element. We can immediately fill out the sides adjacent to the header rows by the group axioms (since $e * g = g$) in Table 1. Therefore we must have that $a * a = e$ and therefore a is its own inverse. We shall also note if we apply the group operation n -times to an element, we can represent it as $a^n = a * a * \dots * a$ (n times). If we take a group of cardinality three, we have that $G = \{e, a, b\}$. We once again

	e	a
e	e	a
a	a	e

Table 1: Group table for a group of cardinality 2.

can fill out the header adjacent row and column by composition with the identity. In the $a * b$ cell of Table 2, we require that be e because otherwise, that would require $a * a$ to be e and therefore $b * a = a$ but a was already in that column and so hence we have a contradiction. Therefore we must have $a * a = b$ and $a * b = e$. From this, we can see that $a^2 = b$ and if we multiply $a * b = e$, this implies $a * a^2 = a^3 = e$. Our group is therefore $G = \{e, a, a^2\}$.

	e	a	b
e	e	a	b
a	a	b	e
b	b	e	a

Table 2: Group table for a group of cardinality 3

Example 1.7. Let us look at the Quaternion group. The Quaternion group is the set

$$Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$$

under standard multiplication. 1 is the identity element and $i^2 = j^2 = k^2 = -1$. Quaternion multiplication is defined as

$$\begin{aligned} ij &= k & jk &= i & ki &= j \\ ji &= -k & kj &= -i & ik &= -j \end{aligned}$$

This is a group of order 8 and is non-abelian. The inverse of -1 is -1 and so because of this we can say the inverse i and $-i$ and same for j and k . Associativity is also good. This group is the reason why the cross product exists in $\mathbb{R}^3$ and the related Octonion group allows for the existence of the cross product in $\mathbb{R}^7$.

Example 1.8. Another important group is the set of complex numbers with norm of 1 under multiplication. In other words, $\{z \in \mathbb{C} : |z| = 1\}$. This is an Abelian group and is of infinite order. This is a group because the norms are all one so multiplication stays within the group ($|z_1||z_2| = |z_1 z_2| = 1$) and the inverse is also in the group (if $|z_1| = 1$, $\frac{1}{|z_1|} = 1$). We can also convert each complex number into its polar form $z = re^{i\theta}$ $r, \theta \in \mathbb{R}$ and since each element has a norm of 1, $r = 1$ and we can just take $z_1 z_2 = e^{i(\theta_1 + \theta_2)}$ and since the real numbers are a group under addition, this set of complex numbers also form a group. This group is also used to generate groups which contain the n^{th} primitive roots of unity. To do this, we fix a number $n \in \mathbb{N}$. Then the number

$$e^{\frac{2k\pi}{n}i} \quad k \in \{1, 2, \dots, n-1\}$$

is an n^{th} primitive root of unity. This means that these are roots to the equation $z^n = 1$ and the primitive means that n is the smallest power that can satisfy the equation (e.g., no smaller number (let's call m) exists such that $z^m = 1$). We then say that the set of the n^{th} groups of unity form an Abelian group under multiplication of order n . One can see this because multiplication of complex numbers is associative, the identity is 1 ($k = 0$ root)

Definition 1.9. Suppose A is any set. A permutation is a function $f : A \mapsto A$ that is bijective.

Proposition 1.10. *The set of all permutations of a set A is a group where the binary operation $*$ is composition and e is the identity permutation.*

Proof. We must first show that the composition of permutations is also a permutation. Let us say that σ, τ are permutations of A . We can now prove that $\sigma\tau$ is injective. Suppose

$$(\sigma\tau)(a) = (\sigma\tau)(b) \rightarrow \sigma(\tau(a)) = \sigma(\tau(b))$$

Since σ is a permutation, σ is injective and therefore

$$\tau(a) = \tau(b)$$

Using that logic again, τ is a permutation and therefore injective and

$$a = b$$

To show that this is onto, we have to show that for some $a \in A$, there exists an $x \in A$ such that $a = \sigma\tau(x)$. Since σ is onto, there exists a $y \in A$ such that $a = \sigma(y)$. τ is also surjective and so we can say there exists an $x \in A$ such that $y = \tau(x)$. Putting everything together we have that

$$a = \sigma(y) = \sigma(\tau(x)) = (\sigma\tau)(x)$$

As such we have shown that $\sigma\tau$ is a permutation. Now we can check the group axioms. The group operation is associative since function is associative

$$((\sigma\tau)\rho)(a) = \sigma(\tau(\rho(a))) = \sigma(\tau\rho(a)) = (\sigma(\tau\rho))(a)$$

We also denote the identity permutation ι such that $\iota(a) = a \ \forall a \in A$. Finally, any permutation is invertible since it is surjective and given any $y \in A$, there exists only one x such that $y = \sigma(x)$ since σ is bijective. We therefore have that $\sigma^{-1}(y) = x$. We can check that σ^{-1} is a permutation by using the fact that $\sigma(\sigma^{-1}(x)) = x$. Let us consider $\sigma^{-1}(y) = \sigma^{-1}(x)$ and applying σ to both sides we get $x = y$ and hence injectivity. We shall define a $y = \sigma(x)$ and therefore $\sigma^{-1}(y) = \sigma^{-1}(\sigma(x)) = x$ for all $y \in A$. $\square$

This group is known as the symmetric group and we denote it S_A . The most important symmetric group is when $A = \{1, 2, \dots, n\}$. We denote this group as S_n and it has cardinality $|S_n| = n!$.

From now on, we will write $a * b$ as ab except when working with groups using $\mathbb{Z}, \mathbb{R}$, etc. with addition as they will be denoted using the usual addition sign. Instead of denoting the identity as e , we shall write 1 under the same condition as the last where there is no confusion is possible.

1.4 The Symmetric Group

Recall that the Symmetric Group S_n is defined as the set of all permutations of a set $\{1, 2, \dots, n\}$ where the binary operation is composition. The cardinality of S_n is $|S_n| = n!$. Elements of S_n can be written as

$$\begin{pmatrix} 1 & 2 & 3 & 4 & \dots & n \\ k_1 & k_2 & k_3 & k_4 & \dots & k_n \end{pmatrix} \text{ where } k_1, \dots, k_n \in \{1, \dots, n\}$$

Since we are using composition, we do the permutation τ before σ in the permutation $\sigma\tau$. Let us look at an example to demonstrate

Example 1.11. Let us suppose in S_5

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 1 & 2 & 3 & 4 \end{pmatrix} \quad \tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 2 & 5 & 4 \end{pmatrix}$$

The permutation $\sigma\tau$ is then

$$\sigma\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 1 & 2 & 4 & 3 \end{pmatrix}$$

We can also find the inverse of a permutation by just going in “reverse”

$$\sigma^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 4 & 5 & 1 \end{pmatrix}$$

Proposition 1.12. *If $n \geq 3$, the group S_n is non-Abelian*

Proof. Let us first just consider the case $n = 3$. Consider the following composition of permutations

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$$

We can prove this for higher n by simply sending everything beyond 3 to their number. For example

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & \dots \\ 2 & 3 & 1 & 4 & 5 & \dots \end{pmatrix}$$

□

1.4.1 Cycle Notation

We can express the elements of a group in a more streamlined way.

$$(k_1, k_2, \dots, k_l) \text{ where } k_1, \dots, k_l \in \{1, 2, \dots, n\} \text{ and are distinct}$$

This notation denotes the following permutation: $k_1 \rightarrow k_2 \rightarrow \dots \rightarrow k_{l-1} \rightarrow k_l \rightarrow k_1$. Any numbers not listed in this cycle is assumed to be fixed. In this case we have to specify what symmetric group we are in because the number of fixed entries could be 1 or 2 or 500.

Example 1.13. In S_6 , the notation $(3, 1, 5, 4)$ represents the permutation

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 2 & 1 & 3 & 4 & 6 \end{pmatrix}$$

We can also compose them together as normal

$$(3, 1, 5, 4)(1, 2, 6, 5, 3) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 6 & 5 & 3 & 1 & 4 \end{pmatrix}$$

If we want to invert a permutation, we just reverse the order of the terms. For example $\sigma = (k_1, k_2, \dots, k_l)$ inverts to $\sigma^{-1} = (k_l, k_{l-1}, \dots, k_1)$. Cycles of length 2 are specifically called *transpositions*. We also note that cycles are not unique since we see that the inverse of a cycle (a, b) is (b, a) which is the same as (a, b) . There is a fact that if you have 2 cycles are disjoint, that is, they don't have any entries in common, the disjoint cycles commute. Additionally, every permutation can be written as a product of disjoint cycles.

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 8 & 5 & 3 & 6 & 7 & 4 & 2 & 1 \end{pmatrix} = (1, 8)(2, 5, 7)(3)(4, 6)$$

The cycles with just a single element can be excluded.

1.5 The Dihedral Group

Start with a regular n -gon. The dihedral group D_n is the group of all symmetries of this n -gon. Let us take a look at the symmetries of a triangle.

Example 1.14. We can consider the symmetries of triangle as a clockwise rotation by 120° once or twice (which is the same as a counter-clockwise rotation by 120°) or reflecting along an axis of symmetry that goes through one of the vertices. These are described in Figure 1. We can call one clockwise rotation by ρ_1 , two clockwise (or again,

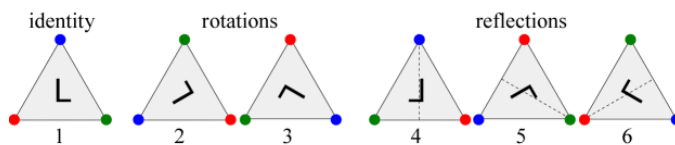


Figure 1: Elements of the Dihedral Group D_3

1 counter-clockwise rotation) by ρ_2 , and then we have reflections μ_1, μ_2, μ_3 about the vertices 1, 2, and 3. We also note that $\rho_1^2 = \rho_2 = 1$ and $\mu_1^2 = \mu_2^2 = \mu_3^2 = 1$. We claim that $D_3 = \{1, \rho_1, \rho_2, \mu_1, \mu_2, \mu_3\}$ are all the elements of the dihedral group D_3 . What about products though? We can think of dihedral group as a subset of the symmetric group and so the symmetries of the n -gon are just permutations of the vertices. However, we need to preserve the relationships between the vertices (basically, there must exist an edge between 1 and 2 and symmetries that don't respect this edge, aren't part of the dihedral group). As such, we can write ρ_1 and μ_2 as

$$\rho_1 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \quad \mu_2 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

We can then find the products of these two actions

$$\rho_1 \mu_2 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} = \mu_1 \quad \mu_2 \rho_1 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} = \mu_3$$

We see that the product of any two symmetries is also a symmetry. We also see the D_n is non-Abelian (which makes sense because we established previously in Proposition 1.12).

Proposition 1.15. *The cardinality of the dihedral group D_n is $|D_n| = 2n$.*

Proof. We can express any rotation as sending the vertex 1 to any vertex $i \in \{1, \dots, n\}$. Vertex 2 can then only be sent to $i - 1$ or $i + 1$ (which come about from reflection). Once these two are established, the rest of the vertices are completely determined due to the “rigid” edge requirement.

Let ρ be a single rotation clockwise by an angle $\frac{2\pi}{n}$, then there are n distinct rotations $\{1, \rho, \rho^2, \dots, \rho^{n-1}\}$ (where $\rho^n = 1$). We can let μ be a reflection about an axis containing the center and vertex 1. We can then achieve any reflection by rotating a certain amount of times between 0 and $n - 1$ and then reflection. Our Dihedral group can then generalized to be $D_n = \{1, \rho, \dots, \rho^{n-1}, \mu, \mu\rho, \dots, \mu\rho^{n-1}\}$. Since there are n number of rotations and n vertices we can reflect about, we have $2n$ elements in our group.

We must also check that all elements are distinct. Each rotation is distinct since if not, then there exists two numbers $i, j \in \{0, \dots, n\}$ such that $\rho^i = \rho^j \rightarrow \rho^{i-j} = e$ which isn't true unless $i = n$ and $j = 0$ since that is the only way to get back the identity permutation. For the reflection permutations, this would imply $\mu\rho^i = \mu\rho^j \rightarrow \rho^i = \rho^j$ for some $i, j \in \{0, \dots, n\}$ which we established isn't possible. $\square$

Proposition 1.16. $\rho\mu = \mu\rho^{-1}$

Proof.

$$\begin{aligned} \rho &= \begin{pmatrix} 1 & 2 & 3 & \dots & n-2 & n-1 & n \\ 2 & 3 & 4 & \dots & n-1 & n & 1 \end{pmatrix} & \rho^{-1} &= \begin{pmatrix} 1 & 2 & 3 & \dots & n-2 & n-1 & n \\ n & 1 & 2 & \dots & n-3 & n-2 & n-1 \end{pmatrix} \\ \mu &= \begin{pmatrix} 1 & 2 & 3 & \dots & n-2 & n-1 & n \\ 1 & n & n-1 & \dots & 4 & 3 & 2 \end{pmatrix} \\ \rho\mu &= \begin{pmatrix} 1 & 2 & 3 & \dots & n-2 & n-1 & n \\ 2 & 1 & n & \dots & 5 & 4 & 3 \end{pmatrix} & \mu\rho^{-1} &= \begin{pmatrix} 1 & 2 & 3 & \dots & n-2 & n-1 & n \\ 2 & 1 & n & \dots & 5 & 4 & 3 \end{pmatrix} \end{aligned}$$

$\square$

1.6 Subgroups

I am making this a subsection because Tate Darin (who can be contacted at tatedarin28@u.northwestern.edu) said it would be funny if subgroups were a subsection.

Definition 1.17. Suppose G is a group. If H is a subset of G which is also a group under the same group operation defined on G , then H is a *subgroup* of G .

Being a subgroup implies the following

1. 1 (which is the identity in G) $\in H$
2. The group operation needs to be closed on H

3. Inverse closure is also required

Every group G has at least 2 subgroups: the trivial subgroup ($\{1\}$) and G itself. If we want a subgroup that is neither of these, we refer to them as *proper, nontrivial* subgroup.

Example 1.18. If we take $G = \mathbb{Z}$ (if we do not specify which group operation with $\mathbb{Z}$, it is assumed to be addition), then a subgroup would be a group $n\mathbb{Z}$ where $n\mathbb{Z}$ is the set of integers divisible by n . In fact, we can state a theorem about the subgroups of $\mathbb{Z}$.

Theorem 1.19. *Every subgroup of $\mathbb{Z}$ is of the form $n\mathbb{Z}$.*